

Regression is taught under at least five different descriptions, and each one is correct. In one, regression is least squares together with the distribution theory of quadratic forms. In another, it is orthogonal projection onto a subspace, and everything else follows as a corollary. A third starts from likelihood and treats the normal linear model as the Gaussian member of the generalized linear family. A fourth defines regression as estimating a conditional distribution, with linearity one structural assumption among many. A fifth treats it as a data-analytic procedure whose assumptions must be checked, one at a time, against the data.

Most textbooks pick one of these descriptions and stay with it. This book uses all of them, in order, as rungs of a single ladder. Parts I–III set up the classical linear model and its exact theory. After that, *each part relaxes exactly one assumption*, and asks what survives, what must be rebuilt, and what new tools the relaxation requires.

Part	What changes	What we study
I	—	The apparatus: matrices, random vectors, the normal distribution, quadratic forms
II	—	$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, $E(\boldsymbol{\varepsilon}) = \mathbf{0}$, $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}$, as a projection problem
III	adds $\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2\mathbf{I})$	Exact finite-sample inference
IV	the model matrix is rank deficient by construction	Analysis of variance as subspace comparison, multi-way layouts, unbalanced data, designed experiments
V	audits every assumption	What breaks, how to see it, what inference survives
VI	drops unbiasedness	Shrinkage, selection, regularization, $p > n$
VII	drops $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}$	Generalized least squares, general Gauss–Markov, mixed models
VIII	drops normality of $\mathbf{Y}$	Exponential families and generalized linear models
IX	drops the linear predictor	Polynomials, splines, smoothing, additive models
X	drops modelling only the mean	Quantile regression and distributional models

Choices this book makes

Geometry before distribution theory. Least squares is introduced as a nearest-point problem ([Chapter 6](#)) before any sampling theory. Fitted values, residuals and sums of squares are projections. Coefficients are coordinates, and coordinates depend on the basis you choose. This is the only framing in which rank-deficient models need no special treatment.

Estimability is an estimation idea. It appears in Part II, not as a preliminary to the analysis of variance. Putting it there is what allows [Part IV](#) to treat ANOVA as ordinary subspace comparison.

Computation is part of the theory. Conditioning, QR and the singular value decomposition come immediately after least squares ([Chapter 10](#)), because numerical stability and statistical stability are closely linked.

The Bayesian treatment is a thread, not a quarantined chapter. Boxed “Bayesian thread” sections run through Parts II, III, VII and VIII — [Section 7.6](#), [Section 12.6](#), [Section 32.6](#) and [Sections 39.3–39.4](#) — each developing the Bayesian counterpart of the result just proved, and the thread runs on into Part IX, where a roughness penalty turns out to be a normal prior ([Section 43.6](#)).

Departures have a theory. [Part V](#) opens by deriving what omitted variables, wrong covariance, nonnormality, outliers and collinearity actually do to the estimator. Diagnostics and remedies come after that.

Material most regression texts omit. This book adds chapters on bootstrap and permutation inference ([Chapter 23](#)), errors in variables ([Chapter 24](#)), the causal interpretation of coefficients ([Chapter 25](#)), high-dimensional regression ([Chapter 28](#)), missing data ([Chapter 41](#)) and structured additive and distributional regression ([Chapter 44](#), [Chapter 45](#)).

Code

Every figure and every number obtained by computation is produced by a Python script in the book’s `code/` directory, and the scripts check their own claims with assertions. The code in the text is taken from those scripts. On this website each code block is a live cell: press **Run** (or Shift+Enter) to execute it in your browser, edit it, and run it again. Nothing needs to be installed; the first run downloads a Python environment with numpy, scipy, pandas and statsmodels. Cells on a page share their variables, and running a cell first runs the cells above it that have not run yet.

Sources

The coverage of this book was planned against six standard texts: Rencher and Schaalje (2008), Seber and Lee (2003), Christensen (2020), Sen and Srivastava (1990), Agresti (2015) and Fahrmeir et al. (2021). Their chapters were mapped onto a single prerequisite graph, and the order of this book respects that graph. Each chapter’s *Notes and sources* section names the treatments it drew on and the primary literature. The exposition, proofs, examples and exercises are original to this book.

How to Use This Book

Parts and chapters

Parts I–VI (Chapters 1–30) are the classical linear model, complete in itself: estimation, exact inference, designed experiments, diagnostics, and the move from unbiased estimation to regularization. Parts [VII–X](#) (Chapters [31–45](#)) climb the rest of the ladder: correlated errors, nonnormal

responses, nonlinear predictors, and models for entire conditional distributions. All forty-five chapters are here; each one links to the results it uses, so any of them can be entered directly.

Reading paths

- **A first course on linear models.** Chapters 5–9, 11–12, 15, 19–20 and 29. Consult Part I as needed.
- **The theory spine.** Part I, then Chapters 6, 7, 8, 9, 11 and 31. This is the shortest route to a full structural understanding of the linear model.
- **For implementers.** Chapters 5, 6, 10 and 20, Part VI and Chapter 34.
- **A course on generalized models.** Chapters 5, 6 and 11, then Parts VII–X.
- **Smoothing and flexible regression.** Chapters 6, 27, 29 and 30, then Part IX; add Chapter 32 for the mixed-model reading of a penalty and Chapter 34 for generalized additive models.
- **Beyond the conditional mean.** Chapters 12, 21 and 22, then Chapter 45.

Prerequisites

Each chapter opens with a list of the earlier chapters it uses, taken from the prerequisite graph the book was planned against. Part I is written as reference material. Read it quickly the first time and come back to it when a later chapter sends you there.

Exercises

Exercises come at the end of each section in three groups: **A. Check your understanding** (routine: a definition or computation has been understood), **B. Practice** (core: the result is used later in the book) and **C. Going deeper** (hard: may need an idea not in the text). Many have a worked solution, folded under the exercise; try the exercise before opening it.

Notation

Symbol	Meaning
$\mathbf{Y}, \mathbf{y}$	response vector (random, observed), $n \times 1$
$\mathbf{X}$	model matrix, $n \times p$; columns $\mathbf{x}_1, \dots, \mathbf{x}_p$
$\mathbf{X}$	a <i>random</i> vector of regressors, in the population sections 5.1, 6.11 and 14.2–14.5 only
$\boldsymbol{\beta}, \hat{\boldsymbol{\beta}}$	coefficient vector and a least squares estimate
$\boldsymbol{\varepsilon}$	error vector, $\boldsymbol{\varepsilon} = \mathbf{Y} - \mathbf{X}\boldsymbol{\beta}$
$\hat{\mathbf{Y}}, \hat{\boldsymbol{\varepsilon}}$	fitted values $\mathbf{X}\hat{\boldsymbol{\beta}}$ and residuals $\mathbf{Y} - \hat{\mathbf{Y}}$
σ^2	error variance
$\mathbb{R}^n$	n -dimensional real space with $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^\top \mathbf{v}$
$\mathcal{C}(\mathbf{A}), \mathcal{N}(\mathbf{A})$	column space and null space of $\mathbf{A}$
$\mathcal{S}^\perp$	orthogonal complement of a subspace $\mathcal{S}$
$\mathbf{A}^\top, \mathbf{A}^-$	transpose; a generalized inverse ($\mathbf{A}\mathbf{A}^-\mathbf{A} = \mathbf{A}$)
$\mathbf{G}$	a generalized inverse of $\mathbf{X}^\top \mathbf{X}$ (Chapters 6–30); from Chapter 31 on, $\text{Cov}(\mathbf{u})$ in a mixed model
$\mathbf{M}$	orthogonal projection onto $\mathcal{C}(\mathbf{X})$

Symbol	Meaning
$\mathbf{M}_0$	orthogonal projection onto the column space of a reduced model
rank, tr, diag	rank, trace, diagonal matrix
$\mathbf{1}, \mathbf{0}, \mathbf{I}$	vector of ones, zero vector, identity
$\ \cdot\ $	Euclidean norm unless stated otherwise
E, Var, Cov	expectation, variance, covariance (matrix)
$N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$	n -variate normal distribution
$\chi^2(r, \gamma), F(r, s, \gamma)$	noncentral distributions, noncentrality γ
$\mathbf{V}$	error covariance: a known shape in $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{V}$ (Chapter 31), or $\mathbf{V} = \mathbf{ZGZ}^\top + \mathbf{R}$ in a mixed model (Chapters 32–33)
$\mathbf{Z}, \mathbf{u}, \mathbf{G}, \mathbf{R}$	random-effect design matrix, random effects, $\text{Cov}(\mathbf{u})$ and $\text{Cov}(\boldsymbol{\epsilon})$ (Part VII)
$\theta, b(\theta), \phi, V(\mu)$	natural parameter, cumulant function, dispersion and variance function of an exponential dispersion family (Part VIII)
$\eta, g, \mathbf{W}, D$	linear predictor $\mathbf{x}_{(i)}^\top \boldsymbol{\beta}$, link function, working weights and deviance of a generalized linear model (Part VIII)
$\mathbf{R}_i, \mathbf{R}$	working correlation matrix of a cluster (Chapter 40) and missingness indicator array (Chapter 41): local uses of $\mathbf{R}$ — as are the residual projection of Chapter 16 and the constraint matrix of Chapter 17 — each announced where it starts
$\mathbf{B}, \boldsymbol{\gamma}$	matrix of basis functions, $B_{ij} = B_j(x_i)$, and its coefficient vector (Part IX)
κ_j, t_j, K	knots and their number: $\kappa_1 < \cdots < \kappa_K$ are the interior knots in Chapter 42; Chapter 43 writes $t_1 < \cdots < t_K$ for the interior knots and keeps κ for the extended B-spline sequence; Chapter 44 writes κ_l for the knots of a kriging term
$\mathbf{P}, \lambda, \mathbf{S}_\lambda$	penalty matrix, smoothing parameter and smoother matrix of a penalized fit (Chapters 43–45); $\text{tr}(\mathbf{S}_\lambda)$ is its effective degrees of freedom
ρ_τ, τ	the check loss $\rho_\tau(u) = u\{\tau - \mathbf{1}\{u < 0\}\}$ and the quantile level (Chapter 45)

Vectors are columns. Matrices and vectors are set in bold upright type and scalars in italic. A bare “projection” always means an orthogonal projection with respect to the Euclidean inner product. Section 6.9 covers the other kinds.